

MAPS

Multiresolution Adaptive Parameterization of Surfaces

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- **Problem statement:**

- Dense triangular meshes from 3D scanners or MRI are complex to handle (size, structure).
- Issues arise in storage, transmission, and editing of surfaces.

- **Objectives of MAPS:**

- Generate smooth parameterizations of complex surfaces.
- Produce efficient, adaptive, and controlled remeshing.

- **Applications include:**

- Compression, texturing, morphing.
- Multiresolution editing.

Compression

Vertex Removal

- The hierarchy is built by progressively removing vertices.
- Vertices are removed in independent groups to simplify the mesh while preserving its topology.
- **Criteria for vertex removal:**
 - Low connectivity degree (outdegree < 12).
 - Geometric criteria (local area and curvature) using a priority queue.
 - Priority weight $w(x, i)$ for a vertex i :

$$w(\lambda, i) = \lambda \cdot \frac{a(i)}{\max_{j \in P^l} a(j)} + (1 - \lambda) \cdot \frac{\kappa(i)}{\max_{j \in P^l} \kappa(j)}$$

where:

- $a(i)$: area of the 1-neighborhood,
- $\kappa(i) = |\kappa_1| + |\kappa_2|$: curvature at the vertex,
- $\lambda \in [0, 1]$: weighting factor (typically $\lambda = 0.5$).
- Overall complexity: $O(N \log N)$ thanks to priority-based selection.

Projection and Retriangulation

- The vertices of the 1-ring neighbourhood are ordered cyclically.
- The deleted 1-ring is projected onto the plane using conformal maps μ_i :

$$\mu_i(p_j) = r_k e^{i\vartheta_k a}, \quad r_k = \|p_i - p_j\|, \quad \vartheta_k = \sum_{\ell=1}^k \angle(p_{j_{\ell-1}}, p_i, p_{j_\ell}).$$

$$a = \frac{2\pi}{\vartheta_{K_I}}.$$

- When the removed vertex lies on the boundary, the 1-ring is mapped onto a half-disk by setting:

$$a = \frac{\pi}{\vartheta_{K_I}}.$$

- The central vertex p_i is mapped to the origin of the plane, i.e., $\mu_i(p_i) = 0$.
- The hole is then retriangulated using a constrained Delaunay

Projection and Retriangulation — Illustration

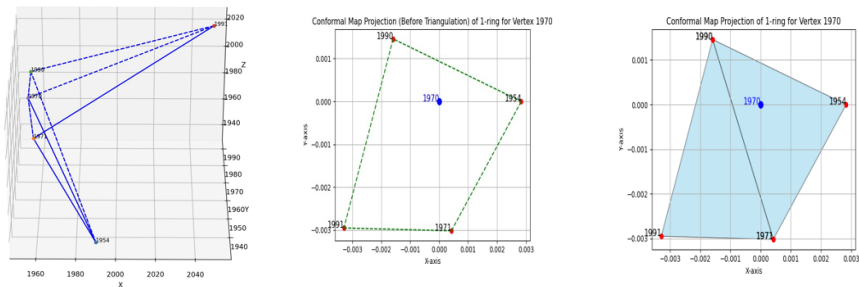


Figure: To remove a vertex p_i , its star neighborhood ($star(i)$) is projected from 3D space onto a plane using the conformal map μ_i . In the plane, the central vertex is deleted and the resulting hole is retriangulated (constrained Delaunay), then lifted back to 3D.

Projection on a Half-Disk (Boundary Case)

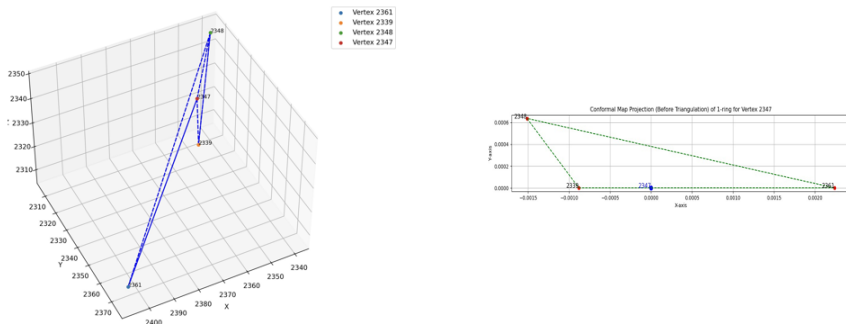


Figure: When the removed vertex p_i lies on the boundary of the mesh, its 1-ring neighbourhood ($star(i)$) is projected onto a **half-disk** instead of a full disk. This ensures that the boundary geometry is preserved during the projection step.

Initial Parameterization : Bijection Π

- To build a parameterization, we start by constructing a bijection Π from $\varphi(|K_L|)$ to $\varphi(|K_0|)$.
- The parameterization of the original mesh over the base domain is given by $\Pi^{-1}(\varphi(|K_0|))$.
- For any point $p \in \varphi(|K_L|)$, its image through Π is $p_0 = \Pi(p) \in \varphi(|K_0|)$, which can be written as:

$$p_0 = \alpha p_i + \beta p_j + \gamma p_k, \quad \{i, j, k\} \in K_0,$$
$$\alpha + \beta + \gamma = 1.$$

- The point p_0 lies inside a triangle of the base domain.

Computing Π_l for the Vertices

- The mapping can be computed simultaneously during the construction of the multiresolution hierarchy.
- The key idea is to compute successive linear bijections Π_l between $\varphi(|K_l|)$ and $\varphi(|K_{l-1}|)$, starting with Π_L (the identity), and ending with the final bijection $\Pi_0 = \Pi$.

Computing Π_{l-1} (Part I)

- Suppose Π_l is known and we want to compute Π_{l-1} . Every vertex $i \in K_l$ falls into one of the following cases:

1. Vertex survives at level $l - 1$.

- If $\{i\} \in K_{l-1}$, the vertex is not removed when going from level l to $l - 1$.
- Then nothing changes:

$$\Pi_{l-1}(p_i) = \Pi_l(p_i) = p_i.$$

2. Vertex is removed at level l .

- Consider the conformal projection of the 1-ring around p_i .
- After retriangulation, the origin lies inside a triangle $t = \{j, k, m\} \in K_{l-1}$.
- Then:

$$\Pi_{l-1}(p_i) = \alpha \mu_i(p_j) + \beta \mu_i(p_k) + \gamma \mu_i(p_m),$$

3. Vertex removed at an earlier level.

- If $\{i\} \in K_L \setminus K_l$, the vertex was removed before level l .
- Then we have:

$$\Pi_l(p_i) = \alpha_0 p_{j_0} + \beta_0 p_{k_0} + \gamma_0 p_{m_0}, \quad t_0 = \{j_0, k_0, m_0\} \in K_l.$$

- If $t_0 \in K_{l-1}$, nothing changes.
- Otherwise, exactly one vertex of t_0 was removed (say j_0).
- Using the conformal map μ_{j_0} and the new retriangulation, we obtain:

$$\Pi_{l-1}(p_i) = \alpha p_j + \beta p_k + \gamma p_m, \quad t = \{j, k, m\} \in K_{l-1}.$$

Base Domain and Mapping through Π

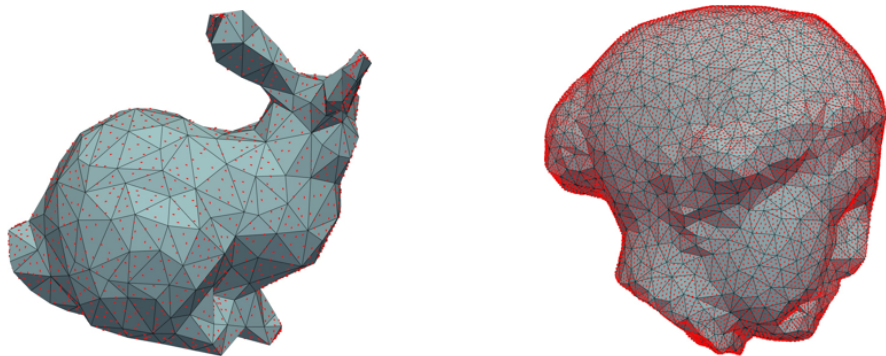


Figure: Base domain $\varphi(|K_0|)$. For each point p_i of the original mesh, its mapped location $\Pi(p_i)$ is shown as a red point on the base domain.

Decompression

Uniform Remeshing

- The base domain is uniformly subdivided using a midpoint subdivision scheme.
- Each newly created point is mapped back onto the original surface through Π^{-1} :

$$q = \alpha \Pi(p_i) + \beta \Pi(p_j) + \gamma \Pi(p_k), \quad \Pi^{-1}(q) = \alpha p_i + \beta p_j + \gamma p_k.$$

- This ensures regular connectivity and faithful reconstruction.

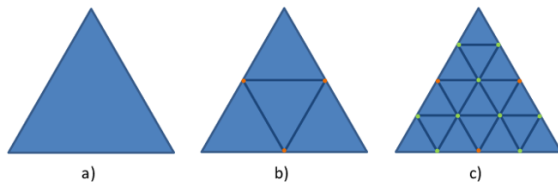


Figure: Midpoint subdivision scheme.

- **Objective:** avoid costly uniform subdivision and focus refinement on localized details.
- The process is guided by the DK hierarchy using an error measure:

$$E(t) = \max_{p_i \in P_L, \Pi(p_i) \in t} \text{dist}(p_i, t),$$

- A triangle is subdivided if

$$\frac{E(t)}{B} > \varepsilon,$$

where B is the longest side of the bounding box and ε is the threshold.

Results

Progressive Reconstruction Results

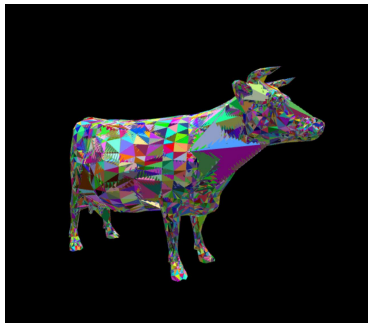
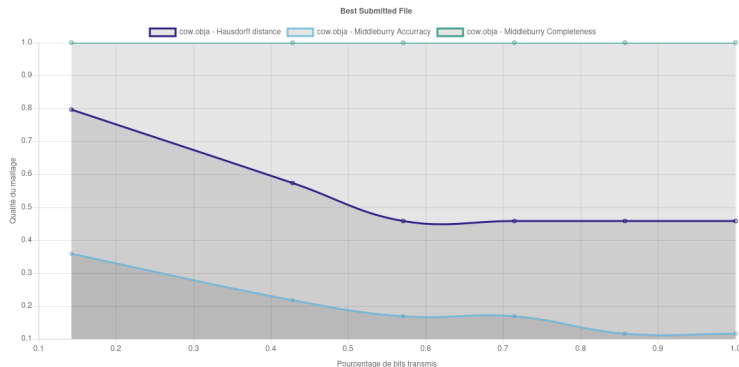


Figure 1: *



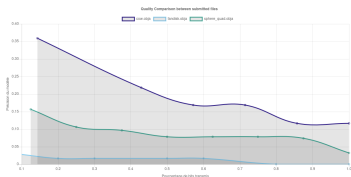
Figure 2: *

Quality Evaluation on the Cow Model

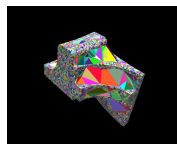


- The reconstruction quality increases as more bits are transmitted.
- Hausdorff distance decreases significantly during the first refinement levels.
- The completeness curve remains stable across all packet levels.
- Accuracy gradually improves before stabilizing at higher resolutions.

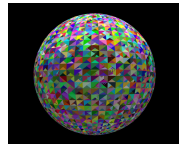
Comparison Across Different Models



*Cow



*Fandisk



*Sphere

Observation:

The cow model exhibits the highest variation in accuracy due to its complex anatomy, while the sphere remains the most stable. The fandisk shows moderate but consistent refinement.